

8 Conformal Mappings

The results I found for polygons can be extended under very general assumptions. I have undertaken this research because it is a step towards a deeper understanding of the mapping problem, for which not much has happened since Riemann's inaugural dissertation; this, even though the theory of mappings, with its close connection with the fundamental theorems of Riemann's function theory, deserves in the highest degree to be developed further.

E. B. Christoffel, 1870

The problems and ideas we present in this chapter are more geometric in nature than the ones we have seen so far. In fact, here we will be primarily interested in mapping properties of holomorphic functions. In particular, most of our results will be “global,” as opposed to the more “local” analytical results proved in the first three chapters. The motivation behind much of our presentation lies in the following simple question:

Given two open sets U and V in $\mathbb{C}$, does there exist a holomorphic bijection between them?

By a holomorphic bijection we simply mean a function that is both holomorphic and bijective. (It will turn out that the inverse map is then automatically holomorphic.) A solution to this problem would permit a transfer of questions about analytic functions from one open set with little geometric structure to another with possibly more useful properties. The prime example consists in taking $V = \mathbb{D}$ the unit disc, where many ideas have been developed to study analytic functions.¹ In fact, since the disc seems to be the most fruitful choice for V we are led to a variant of the above question:

Given an open subset Ω of $\mathbb{C}$, what conditions on Ω guarantee that there exists a holomorphic bijection from Ω to $\mathbb{D}$?

¹For the corresponding problem when $V = \mathbb{C}$, the solution is trivial: only $U = \mathbb{C}$ is possible. See Exercise 14 in Chapter 3.

In some instances when a bijection exists it can be given by explicit formulas, and we turn to this aspect of the theory first. For example, the upper half-plane can be mapped by a holomorphic bijection to the disc, and this is given by a fractional linear transformation. From there, one can construct many other examples, by composing simple maps already encountered earlier, such as rational functions, trigonometric functions, logarithms, etc. As an application, we discuss the consequence of these constructions to the solution of the Dirichlet problem for the Laplacian in some particular domains.

Next, we pass from the specific examples to prove the first general result of the chapter, namely the Schwarz lemma, with an immediate application to the determination of all holomorphic bijections (“automorphisms” of the disc to itself). These are again given by fractional linear transformations.

Then comes the heart of the matter: the Riemann mapping theorem, which states that Ω can be mapped to the unit disc whenever it is simply connected and not all of $\mathbb{C}$. This is a remarkable theorem, since little is assumed about Ω , not even regularity of its boundary $\partial\Omega$. (After all, the boundary of the disc is smooth.) In particular, the interiors of triangles, squares, and in fact any polygon can be mapped via a bijective holomorphic function to the disc. A precise description of the mapping in the case of polygons, called the Schwarz-Christoffel formula, will be taken up in the last section of the chapter. It is interesting to note that the mapping functions for rectangles are given by “elliptic integrals,” and these lead to doubly-periodic functions. The latter are the subject of the next chapter.

1 Conformal equivalence and examples

We fix some terminology that we shall use in the rest of this chapter. A bijective holomorphic function $f : U \rightarrow V$ is called a **conformal map** or **biholomorphism**. Given such a mapping f , we say that U and V are **conformally equivalent** or simply **biholomorphic**. An important fact is that the inverse of f is then automatically holomorphic.

Proposition 1.1 *If $f : U \rightarrow V$ is holomorphic and injective, then $f'(z) \neq 0$ for all $z \in U$. In particular, the inverse of f defined on its range is holomorphic, and thus the inverse of a conformal map is also holomorphic.*

Proof. We argue by contradiction, and suppose that $f'(z_0) = 0$ for some $z_0 \in U$. Then

$$f(z) - f(z_0) = a(z - z_0)^k + G(z) \quad \text{for all } z \text{ near } z_0,$$

with $a \neq 0$, $k \geq 2$ and G vanishing to order $k + 1$ at z_0 . For sufficiently small w , we write

$$f(z) - f(z_0) - w = F(z) + G(z), \quad \text{where } F(z) = a(z - z_0)^k - w.$$

Since $|G(z)| < |F(z)|$ on a small circle centered at z_0 , and F has at least two zeros inside that circle, Rouché's theorem implies that $f(z) - f(z_0) - w$ has at least two zeros there. Since $f'(z) \neq 0$ for all $z \neq z_0$ but sufficiently close to z_0 it follows that the roots of $f(z) - f(z_0) - w$ are distinct, hence f is not injective, a contradiction.

Now let $g = f^{-1}$ denote the inverse of f on its range, which we can assume is V . Suppose $w_0 \in V$ and w is close to w_0 . Write $w = f(z)$ and $w_0 = f(z_0)$. If $w \neq w_0$, we have

$$\frac{g(w) - g(w_0)}{w - w_0} = \frac{1}{\frac{w - w_0}{g(w) - g(w_0)}} = \frac{1}{\frac{f(z) - f(z_0)}{z - z_0}}.$$

Since $f'(z_0) \neq 0$, we may let $z \rightarrow z_0$ and conclude that g is holomorphic at w_0 with $g'(w_0) = 1/f'(g(w_0))$.

From this proposition we conclude that two open sets U and V are conformally equivalent if and only if there exist holomorphic functions $f : U \rightarrow V$ and $g : V \rightarrow U$ such that $g(f(z)) = z$ and $f(g(w)) = w$ for all $z \in U$ and $w \in V$.

We point out that the terminology adopted here is not universal. Some authors call a holomorphic map $f : U \rightarrow V$ conformal if $f'(z) \neq 0$ for all $z \in U$. This definition is clearly less restrictive than ours; for example, $f(z) = z^2$ on the punctured disc $\mathbb{C} - \{0\}$ satisfies $f'(z) \neq 0$, but is not injective. However, the condition $f'(z) \neq 0$ is tantamount to f being a local bijection (Exercise 1). There is a geometric consequence of the condition $f'(z) \neq 0$ and it is at the root of this discrepancy of terminology in the definitions. A holomorphic map that satisfies this condition preserves angles. Loosely speaking, if two curves γ and η intersect at z_0 , and α is the oriented angle between the tangent vectors to these curves, then the image curves $f \circ \gamma$ and $f \circ \eta$ intersect at $f(z_0)$, and their tangent vectors form the same angle α . Problem 2 develops this idea.

We begin our study of conformal mappings by looking at a number of specific examples. The first gives the conformal equivalence between

the unit disc and the upper half-plane, which plays an important role in many problems.

1.1 The disc and upper half-plane

The upper half-plane, which we denote by $\mathbb{H}$, consists of those complex numbers with positive imaginary part; that is,

$$\mathbb{H} = \{z \in \mathbb{C} : \operatorname{Im}(z) > 0\}.$$

A remarkable fact, which at first seems surprising, is that the unbounded set $\mathbb{H}$ is conformally equivalent to the unit disc. Moreover, an explicit formula giving this equivalence exists. Indeed, let

$$F(z) = \frac{i - z}{i + z} \quad \text{and} \quad G(w) = i \frac{1 - w}{1 + w}.$$

Theorem 1.2 *The map $F : \mathbb{H} \rightarrow \mathbb{D}$ is a conformal map with inverse $G : \mathbb{D} \rightarrow \mathbb{H}$.*

Proof. First we observe that both maps are holomorphic in their respective domains. Then we note that any point in the upper half-plane is closer to i than to $-i$, so $|F(z)| < 1$ and F maps $\mathbb{H}$ into $\mathbb{D}$. To prove that G maps into the upper half-plane, we must compute $\operatorname{Im}(G(w))$ for $w \in \mathbb{D}$. To this end we let $w = u + iv$, and note that

$$\begin{aligned} \operatorname{Im}(G(w)) &= \operatorname{Re} \left(\frac{1 - u - iv}{1 + u + iv} \right) \\ &= \operatorname{Re} \left(\frac{(1 - u - iv)(1 + u - iv)}{(1 + u)^2 + v^2} \right) \\ &= \frac{1 - u^2 - v^2}{(1 + u)^2 + v^2} > 0 \end{aligned}$$

since $|w| < 1$. Therefore G maps the unit disc to the upper half-plane. Finally,

$$F(G(w)) = \frac{i - i \frac{1-w}{1+w}}{i + i \frac{1-w}{1+w}} = \frac{1 + w - 1 + w}{1 + w + 1 - w} = w,$$

and similarly $G(F(z)) = z$. This proves the theorem.

An interesting aspect of these functions is their behavior on the boundaries of our open sets.² Observe that F is holomorphic everywhere on $\mathbb{C}$

²The boundary behavior of conformal maps is a recurrent theme that plays an important role in this chapter.

except at $z = -i$, and in particular it is continuous everywhere on the boundary of $\mathbb{H}$, namely the real line. If we take $z = x$ real, then the distance from x to i is the same as the distance from x to $-i$, therefore $|F(x)| = 1$. Thus F maps $\mathbb{R}$ onto the boundary of $\mathbb{D}$. We get more information by writing

$$F(x) = \frac{i-x}{i+x} = \frac{1-x^2}{1+x^2} + i \frac{2x}{1+x^2},$$

and parametrizing the real line by $x = \tan t$ with $t \in (-\pi/2, \pi/2)$. Since

$$\sin 2a = \frac{2 \tan a}{1 + \tan^2 a} \quad \text{and} \quad \cos 2a = \frac{1 - \tan^2 a}{1 + \tan^2 a},$$

we have $F(x) = \cos 2t + i \sin 2t = e^{i2t}$. Hence the image of the real line is the arc consisting of the circle omitting the point -1 . Moreover, as x travels from $-\infty$ to ∞ , $F(x)$ travels along that arc starting from -1 and first going through that part of the circle that lies in the lower half-plane.

The point -1 on the circle corresponds to the “point at infinity” of the upper half-plane.

Remark. Mappings of the form

$$z \mapsto \frac{az + b}{cz + d},$$

where a, b, c , and d are complex numbers, and where the denominator is assumed not to be a multiple of the numerator, are usually referred to as **fractional linear transformations**. Other instances occur as the automorphisms of the disc and of the upper half-plane in Theorems 2.1 and 2.4.

1.2 Further examples

We gather here several illustrations of conformal mappings. In certain cases we discuss the behavior of the map on the boundary of the relevant domain. Some of the mappings are pictured in Figure 1.

EXAMPLE 1. Translations and dilations provide the first simple examples. Indeed, if $h \in \mathbb{C}$, the translation $z \mapsto z + h$ is a conformal map from $\mathbb{C}$ to itself whose inverse is $w \mapsto w - h$. If h is real, then this translation is also a conformal map from the upper half-plane to itself.

For any non-zero complex number c , the map $f : z \mapsto cz$ is a conformal map from the complex plane to itself, whose inverse is simply $g : w \mapsto c^{-1}w$. If c has modulus 1, so that $c = e^{i\varphi}$ for some real φ , then f is